Brutal Practice Paper B31 - Transportation, Simple Equations, Acids Bases & Salts, Exponents & Powers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A resting heart beats 72 times every minute; keeping that steady rate, how many times will it beat during one complete hour? [Transportation in Animals & Plants]

- **(A)** 432 beats
- **(B)** 4320 beats
- **(C)** 1320 beats
- **(D)** 132 beats
- **(E)** 720 beats
- **(F)** 2160 beats
- **(G)** 4200 beats

2. Three times a number, decreased by 7, equals twice the number increased by 5. What is the number? [Simple Equations]

- **(A)** 12.5
- **(B)** 2
- **(C)** -2
- **(D)** 6
- **(E)** -12
- **(F)** 12

3. Exactly 25 mL of an acid is neutralised by 25 mL of a base. How much of the same base is needed to neutralise 100 mL of that acid? [Acids, Bases & Salts]

- **(A)** 50 mL
- **(B)** 400 mL
- **(C)** 75 mL
- **(D)** 100 mL
- **(E)** 10 mL
- **(F)** 125 mL
- **(G)** 25 mL

4. A single cell divides into 2 every hour. Starting from 1 cell, how many cells are there after 6 hours? [Exponents & Powers]

- **(A)** 48
- **(B)** 32
- **(C)** 36
- **(D)** 64
- **(E)** 6
- **(F)** 12
- **(G)** 128

5. A nurse counts 18 pulse beats in 15 seconds. What is the heart rate per minute, and is it inside the normal 60-100 range? [Transportation in Animals & Plants]

- **(A)** 60 per minute, normal
- **(B)** 108 per minute, high
- **(C)** 72 per minute, normal
- **(D)** 36 per minute, low
- **(E)** 270 per minute, high
- **(F)** 90 per minute, normal

6. The sum of three consecutive whole numbers is 96. What is the largest of them? [Simple Equations]

- **(A)** 34
- **(B)** 30
- **(C)** 96
- **(D)** 31
- **(E)** 48
- **(F)** 33

7. One antacid tablet neutralises 5 mL of stomach acid. How many tablets are needed to neutralise 35 mL of acid? [Acids, Bases & Salts]

- **(A)** 8 tablets
- **(B)** 175 tablets
- **(C)** 30 tablets
- **(D)** 6 tablets
- **(E)** 5 tablets
- **(F)** 40 tablets
- **(G)** 7 tablets

8. Simplify 2^5 x 2^3 / 2^4 and give its value. [Exponents & Powers]

- **(A)** 2
- **(B)** 256
- **(C)** 16
- **(D)** 64
- **(E)** 32
- **(F)** 8
- **(G)** 12

9. A heart pumps about 70 mL of blood with each beat and beats 72 times a minute. How many litres does it pump in one minute? [Transportation in Animals & Plants]

- **(A)** 5.04 L
- **(B)** 0.504 L
- **(C)** 142 L
- **(D)** 504 L
- **(E)** 4.2 L
- **(F)** 5040 L

10. A number multiplied by 5, then increased by 8, equals 8 less than 7 times the number. Find the number. [Simple Equations]

- **(A)** 4
- **(B)** 8
- **(C)** 2
- **(D)** 1
- **(E)** 0
- **(F)** -8
- **(G)** 16

11. Vinegar is 5% acetic acid by volume. How much pure acetic acid is in a 250 mL bottle of vinegar? [Acids, Bases & Salts]

- **(A)** 12.5 mL
- **(B)** 25 mL
- **(C)** 237.5 mL
- **(D)** 50 mL
- **(E)** 125 mL
- **(F)** 1.25 mL

12. Which whole number, raised to the power 3, gives 343? [Exponents & Powers]

- **(A)** 9
- **(B)** 17
- **(C)** 3
- **(D)** 49
- **(E)** 7
- **(F)** 6
- **(G)** 21

13. A plant takes up 5 L of water in a day and transpires 99% of it. How much water, in millilitres, is kept for the plant's own use? [Transportation in Animals & Plants]

- **(A)** 100 mL
- **(B)** 50 mL
- **(C)** 4950 mL
- **(D)** 5 mL
- **(E)** 250 mL
- **(F)** 4950 L

14. A mother is 5 times as old as her daughter. In 6 years she will be only 3 times as old. How old is the daughter now? [Simple Equations]

- **(A)** 5 years
- **(B)** 18 years
- **(C)** 8 years
- **(D)** 30 years
- **(E)** 12 years
- **(F)** 6 years

15. If 8 mL of base just neutralises 10 mL of an acid, how much base is needed for 40 mL of the same acid? [Acids, Bases & Salts]

- **(A)** 42 mL
- **(B)** 48 mL
- **(C)** 50 mL
- **(D)** 32 mL
- **(E)** 40 mL
- **(F)** 320 mL
- **(G)** 20 mL

16. Find the value of (3^2) raised to the power 3. [Exponents & Powers]

- **(A)** 27
- **(B)** 216
- **(C)** 729
- **(D)** 9
- **(E)** 243
- **(F)** 81
- **(G)** 18

17. Through tiny pores a leaf transpires 250 mL of water each hour for 8 sunlit hours. What total volume of water does it lose that day? [Transportation in Animals & Plants]

- **(A)** 20 L
- **(B)** 258 mL
- **(C)** 16 L
- **(D)** 2 mL
- **(E)** 0.2 L
- **(F)** 2000 L
- **(G)** 2 L

18. One-third of a number added to one-fourth of the same number gives 14. What is the number? [Simple Equations]

- **(A)** 168
- **(B)** 7
- **(C)** 24
- **(D)** 12
- **(E)** 48
- **(F)** 2
- **(G)** 14

19. An acidic field needs 4 kg of lime, a base, per 100 square metres. How much lime is needed for 2500 square metres? [Acids, Bases & Salts]

- **(A)** 10 kg
- **(B)** 1000 kg
- **(C)** 625 kg
- **(D)** 250 kg
- **(E)** 100 kg
- **(F)** 40 kg

20. Compute the value of 2^3 x 3^2. [Exponents & Powers]

- **(A)** 36
- **(B)** 216
- **(C)** 72
- **(D)** 100
- **(E)** 30
- **(F)** 48
- **(G)** 64

21. Blood is about 7% of body mass. For a person of mass 50 kg, what is the mass of blood, in kilograms? [Transportation in Animals & Plants]

- **(A)** 14 kg
- **(B)** 3.5 g
- **(C)** 7 kg
- **(D)** 35 kg
- **(E)** 0.35 kg
- **(F)** 3.5 kg
- **(G)** 5 kg

22. A rectangle's length is 3 cm more than twice its width, and its perimeter is 36 cm. Find the width. [Simple Equations]

- **(A)** 13 cm
- **(B)** 6 cm
- **(C)** 10 cm
- **(D)** 9 cm
- **(E)** 8 cm
- **(F)** 5.5 cm
- **(G)** 5 cm

23. A bottle is labelled 36% hydrochloric acid by mass. How many grams of acid are in 50 g of this solution? [Acids, Bases & Salts]

- **(A)** 50 g
- **(B)** 14 g
- **(C)** 32 g
- **(D)** 18 g
- **(E)** 9 g
- **(F)** 36 g
- **(G)** 1.8 g

24. Simplify 5^6 / 5^4 and give the number. [Exponents & Powers]

- **(A)** 625
- **(B)** 10
- **(C)** 25
- **(D)** 5
- **(E)** 1
- **(F)** 20

25. In 5 L of blood, plasma is 55% by volume. How many millilitres are blood cells? [Transportation in Animals & Plants]

- **(A)** 2250 mL
- **(B)** 2750 mL
- **(C)** 275 mL
- **(D)** 250 mL
- **(E)** 2200 mL
- **(F)** 2250 L

26. When 5 is added to twice a number, the result equals 4 subtracted from three times the number. Find the number. [Simple Equations]

- **(A)** 1
- **(B)** -9
- **(C)** 4.5
- **(D)** 18
- **(E)** 9
- **(F)** 3

27. If 3 mL of base neutralises 2 mL of acid, how much base neutralises 30 mL of the same acid? [Acids, Bases & Salts]

- **(A)** 45 mL
- **(B)** 35 mL
- **(C)** 30 mL
- **(D)** 20 mL
- **(E)** 50 mL
- **(F)** 15 mL

28. What is the value of 10^4 + 10^3? [Exponents & Powers]

- **(A)** 100000
- **(B)** 1100
- **(C)** 10010
- **(D)** 2000
- **(E)** 11000
- **(F)** 10100

29. A dialysis patient has the blood cleaned 3 times a week. Over one year of 52 weeks, how many sessions is that? [Transportation in Animals & Plants]

- **(A)** 312 sessions
- **(B)** 104 sessions
- **(C)** 49 sessions
- **(D)** 55 sessions
- **(E)** 1095 sessions
- **(F)** 156 sessions

30. Two numbers add up to 60, and one is twice the other. What is the larger number? [Simple Equations]

- **(A)** 45
- **(B)** 30
- **(C)** 20
- **(D)** 50
- **(E)** 40
- **(F)** 15

31. From this list - lemon juice, soap solution, vinegar, baking-soda solution, curd, lime water - how many are acidic? [Acids, Bases & Salts]

- **(A)** 2
- **(B)** 3
- **(C)** 1
- **(D)** 0
- **(E)** 5
- **(F)** 6
- **(G)** 4

32. Find the value of (-2)^4 + (-2)^3. [Exponents & Powers]

- **(A)** -16
- **(B)** 16
- **(C)** -24
- **(D)** 24
- **(E)** -8
- **(F)** 0
- **(G)** 8

33. A leaf has 320 stomata per square millimetre. Its lower surface is 5 square centimetres. How many stomata are there in all? [Transportation in Animals & Plants]

- **(A)** 800000 stomata
- **(B)** 325 stomata
- **(C)** 32000 stomata
- **(D)** 16000 stomata
- **(E)** 1600 stomata
- **(F)** 160000 stomata

34. The sum of three consecutive even numbers is 84. What is the middle number? [Simple Equations]

- **(A)** 84
- **(B)** 30
- **(C)** 24
- **(D)** 42
- **(E)** 27
- **(F)** 26
- **(G)** 28

35. A solution has pH 9 and is tested with litmus. Which statement about its nature and colour change is correct? [Acids, Bases & Salts]

(A) Basic; turns red litmus blue
(B) Basic; turns litmus colourless
(C) Acidic; turns red litmus blue
(D) Basic; turns blue litmus red
(E) Acidic; turns blue litmus red
(F) Neutral; turns litmus green

36. Write 84000 in standard form. [Exponents & Powers]

(A) 8.4 x 10^5
(B) 8.4 x 10^2
(C) 0.84 x 10^5
(D) 8.4 x 10^4
(E) 84 x 10^4
(F) 8.4 x 10^3

37. Sap rises in a tall tree at 1.5 m per hour. How long will it take to reach a leaf 12 m above the roots? [Transportation in Animals & Plants]

(A) 18 hours
(B) 13.5 hours
(C) 10.5 hours
(D) 8 hours
(E) 6 hours
(F) 80 hours
(G) 16 hours

38. If 7 is subtracted from 4 times a number, the result is 25. Find the number. [Simple Equations]

(A) 7
(B) 8
(C) 32
(D) 9
(E) 4.5
(F) 18

39. Lemon juice has pH 2 and pure water has pH 7. By how many pH units do they differ? [Acids, Bases & Salts]

(A) 4 units
(B) 2 units
(C) 9 units
(D) 7 units
(E) 3 units
(F) 5 units
(G) 14 units

40. Simplify 3^4 / 3^2 x 3 and give the value. [Exponents & Powers]

- **(A)** 27
- **(B)** 243
- **(C)** 81
- **(D)** 18
- **(E)** 3
- **(F)** 12
- **(G)** 9

41. A blood-pressure reading is 120/80 mm Hg. The pulse pressure is the difference between the two numbers. What is it? [Transportation in Animals & Plants]

- **(A)** 40 mm Hg
- **(B)** 160 mm Hg
- **(C)** 200 mm Hg
- **(D)** 60 mm Hg
- **(E)** 1.5 mm Hg
- **(F)** 20 mm Hg
- **(G)** 100 mm Hg

42. A piggy bank has only Rs 5 and Rs 2 coins, 30 coins in all, worth Rs 99. How many Rs 5 coins are there? [Simple Equations]

- **(A)** 15
- **(B)** 17
- **(C)** 20
- **(D)** 10
- **(E)** 12
- **(F)** 13
- **(G)** 18

43. A tank's pH must be 7 but it reads 4. Each spoon of base raises the pH by 0.5. How many spoons are needed? [Acids, Bases & Salts]

- **(A)** 4 spoons
- **(B)** 1.5 spoons
- **(C)** 5 spoons
- **(D)** 3 spoons
- **(E)** 12 spoons
- **(F)** 6 spoons
- **(G)** 7 spoons

44. Between the two powers 2^5 and 5^2, state which one is the larger and by how much. [Exponents & Powers]

- **(A)** 5^2, by 10
- **(B)** 2^5, by 7
- **(C)** 5^2, by 7
- **(D)** equal
- **(E)** 2^5, by 57
- **(F)** 2^5, by 10
- **(G)** 2^5, by 3

45. A person donates 350 mL of blood from a total of about 5000 mL. What percentage of the blood is given? [Transportation in Animals & Plants]

- **(A)** 35%
- **(B)** 3.5%
- **(C)** 0.7%
- **(D)** 14%
- **(E)** 5%
- **(F)** 70%
- **(G)** 7%

46. Two-fifths of a number is 18. What is one-third of the same number? [Simple Equations]

- **(A)** 9
- **(B)** 15
- **(C)** 7.2
- **(D)** 30
- **(E)** 12
- **(F)** 45
- **(G)** 6

47. An ant bite injects 0.2 mL of formic acid. Baking soda neutralises it at 3 mL of base per 1 mL of acid. How much baking-soda solution is needed? [Acids, Bases & Salts]

- **(A)** 3.2 mL
- **(B)** 0.2 mL
- **(C)** 0.6 mL
- **(D)** 0.06 mL
- **(E)** 1.5 mL
- **(F)** 0.4 mL
- **(G)** 6 mL

48. A sheet of paper is folded in half repeatedly. After 10 folds, into how many layers is it divided? [Exponents & Powers]

- **(A)** 20
- **(B)** 256
- **(C)** 100
- **(D)** 210
- **(E)** 512
- **(F)** 1024

49. After training, a runner's resting heart rate falls from 80 to 60 beats per minute. What is the percentage decrease? [Transportation in Animals & Plants]

- **(A)** 33.3%
- **(B)** 20%
- **(C)** 25%
- **(D)** 15%
- **(E)** 75%
- **(F)** 40%

50. A number is divided by 4 and then 3 is added, giving 11. What is the number? [Simple Equations]

- **(A)** 20
- **(B)** 32
- **(C)** 2
- **(D)** 44
- **(E)** 35
- **(F)** 8
- **(G)** 56

51. A salt forms when 40 g of acid reacts fully with 30 g of base. What mass of products forms, given that mass is conserved? [Acids, Bases & Salts]

- **(A)** 10 g
- **(B)** 100 g
- **(C)** 70 g
- **(D)** 35 g
- **(E)** 40 g
- **(F)** 30 g
- **(G)** 1200 g

52. Find the value of 2^0 + 2^1 + 2^2 + 2^3. [Exponents & Powers]

- **(A)** 14
- **(B)** 30
- **(C)** 15
- **(D)** 16
- **(E)** 8
- **(F)** 7

53. A tree must move about 200 kg of water to build 1 kg of dry plant material. How much water is needed to build 6 kg of dry material? [Transportation in Animals & Plants]

- **(A)** 600 kg
- **(B)** 206 kg
- **(C)** 1200 g
- **(D)** 2000 kg
- **(E)** 1200 kg
- **(F)** 12000 kg
- **(G)** 120 kg

54. A is twice as old as B. Five years ago, the sum of their ages was 20. How old is A now? [Simple Equations]

- **(A)** 18 years
- **(B)** 25 years
- **(C)** 20 years
- **(D)** 10 years
- **(E)** 15 years
- **(F)** 30 years
- **(G)** 12 years

55. Rain is acid rain when its pH is below 5.6. Readings are 6.1, 5.0, 4.3, 6.5 and 5.5. How many count as acid rain? [Acids, Bases & Salts]

- **(A)** 6
- **(B)** 4
- **(C)** 5
- **(D)** 1
- **(E)** 0
- **(F)** 3
- **(G)** 2

56. If a^3 x a^2 is worked out with a = 2, what is the result? [Exponents & Powers]

- **(A)** 10
- **(B)** 36
- **(C)** 25
- **(D)** 32
- **(E)** 64
- **(F)** 8
- **(G)** 16

57. During exercise the heart rate rises from 75 to 150 beats per minute. By what percentage has it increased? [Transportation in Animals & Plants]

- **(A)** 200%
- **(B)** 25%
- **(C)** 150%
- **(D)** 2%
- **(E)** 100%
- **(F)** 50%

58. A number increased by 60% of itself becomes 64. What is the number? [Simple Equations]

- **(A)** 48
- **(B)** 38.4
- **(C)** 26
- **(D)** 64
- **(E)** 40
- **(F)** 32
- **(G)** 100

59. Toothpaste is basic and neutralises mouth acid. To take acid of pH 5 up to pH 7 when each brush raises pH by 0.4, how many brushings-worth is needed? [Acids, Bases & Salts]

- **(A)** 4
- **(B)** 6
- **(C)** 8
- **(D)** 0.8
- **(E)** 2
- **(F)** 5

60. Express ten lakh (1000000) as a power of 10. [Exponents & Powers]

- **(A)** 10^6
- **(B)** 10^8
- **(C)** 100^3
- **(D)** 6^10
- **(E)** 10^4
- **(F)** 10^5
- **(G)** 10^7

61. An adult breathes about 15 times a minute while the heart beats 75 times a minute. How many heartbeats occur per single breath? [Transportation in Animals & Plants]

- **(A)** 5 beats
- **(B)** 60 beats
- **(C)** 4 beats
- **(D)** 1125 beats
- **(E)** 15 beats
- **(F)** 90 beats
- **(G)** 45 beats

62. The length of a rectangle is twice its breadth, and its perimeter is 48 cm. What is its area? [Simple Equations]

- **(A)** 64 cm^2
- **(B)** 24 cm^2
- **(C)** 96 cm^2
- **(D)** 128 cm^2
- **(E)** 256 cm^2
- **(F)** 144 cm^2
- **(G)** 48 cm^2

63. China rose indicator turns acids pink and bases green. Across 8 samples it shows pink 5 times. How many samples were basic, assuming none are neutral? [Acids, Bases & Salts]

- **(A)** 13
- **(B)** 1
- **(C)** 4
- **(D)** 3
- **(E)** 8
- **(F)** 2

64. Find the value of 7^0 + 3^0 + 2^2. [Exponents & Powers]

- **(A)** 2
- **(B)** 6
- **(C)** 0
- **(D)** 4
- **(E)** 5
- **(F)** 11
- **(G)** 8

65. Red blood cells live about 120 days before being replaced. What fraction of them is replaced in a single day? [Transportation in Animals & Plants]

- **(A)** 1/120
- **(B)** 1/10
- **(C)** 1/12
- **(D)** 1/60
- **(E)** 1/365
- **(F)** 1/24
- **(G)** 1/100

66. The angles of a triangle are in the ratio 1 : 2 : 3. What is the largest angle? [Simple Equations]

- **(A)** 180 degrees
- **(B)** 90 degrees
- **(C)** 30 degrees
- **(D)** 120 degrees
- **(E)** 100 degrees
- **(F)** 60 degrees

67. Milk of magnesia, a base, neutralises stomach acid; one dose handles 12 mL of acid. If a patient makes 96 mL of excess acid in a day, how many doses are needed? [Acids, Bases & Salts]

- **(A)** 7 doses
- **(B)** 8 doses
- **(C)** 12 doses
- **(D)** 10 doses
- **(E)** 1152 doses
- **(F)** 6 doses
- **(G)** 84 doses

68. Simplify 4^3 / 2^4 to a single number. [Exponents & Powers]

- **(A)** 16
- **(B)** 2
- **(C)** 1
- **(D)** 32
- **(E)** 64
- **(F)** 4
- **(G)** 8

69. A patient's drip delivers 2.5 mL of fluid each minute. How many millilitres enter the blood in 2 hours? [Transportation in Animals & Plants]

- **(A)** 30 mL
- **(B)** 3000 mL
- **(C)** 5 mL
- **(D)** 150 mL
- **(E)** 120 mL
- **(F)** 300 mL
- **(G)** 250 mL

70. Solve for the number: three times the quantity (the number minus 2) equals two times the quantity (the number plus 4). [Simple Equations]

- **(A)** 4
- **(B)** -2
- **(C)** 7
- **(D)** 14
- **(E)** -14
- **(F)** 10

71. Vinegar (acid) and baking soda (base) react in a 5 mL to 4 g ratio. How many grams of baking soda react fully with 35 mL of vinegar? [Acids, Bases & Salts]

- **(A)** 175 g
- **(B)** 7 g
- **(C)** 25 g
- **(D)** 32 g
- **(E)** 44 g
- **(F)** 28 g

72. A microbe triples every hour. Starting from 1, how many are there after 4 hours? [Exponents & Powers]

- **(A)** 9
- **(B)** 12
- **(C)** 64
- **(D)** 43
- **(E)** 243
- **(F)** 81
- **(G)** 27

73. Blood takes about 1 minute to make one full round of the body. How many complete rounds does it make in one full day? [Transportation in Animals & Plants]

- **(A)** 60 rounds
- **(B)** 720 rounds
- **(C)** 144 rounds
- **(D)** 24 rounds
- **(E)** 1440 rounds
- **(F)** 1440 hours
- **(G)** 86400 rounds

74. A man spends half his monthly income on rent and a quarter on food, then saves the remaining Rs 5000. What is his income? [Simple Equations]

- **(A)** Rs 35000
- **(B)** Rs 25000
- **(C)** Rs 40000
- **(D)** Rs 10000
- **(E)** Rs 20000
- **(F)** Rs 5000

75. Three solutions read pH 3, pH 7 and pH 11. Matching them in that order to acid, neutral or base, which is correct? [Acids, Bases & Salts]

- **(A)** base, acid, neutral
- **(B)** acid, neutral, base
- **(C)** acid, base, neutral
- **(D)** neutral, acid, base
- **(E)** acid, neutral, acid
- **(F)** neutral, neutral, base

76. Simplify (2^2)^3 x 2 and give the value. [Exponents & Powers]

- **(A)** 128
- **(B)** 16
- **(C)** 64
- **(D)** 32
- **(E)** 48
- **(F)** 12

77. Of every 100 mL of blood, about 45 mL is red cells. In a 4 L blood sample, what volume is red cells? [Transportation in Animals & Plants]

- **(A)** 18000 mL
- **(B)** 180 mL
- **(C)** 45 mL
- **(D)** 2200 mL
- **(E)** 1800 mL
- **(F)** 450 mL
- **(G)** 1.8 mL

78. Five years from now a girl will be three times as old as she was five years ago. How old is she now? [Simple Equations]

- **(A)** 20 years
- **(B)** 15 years
- **(C)** 10 years
- **(D)** 7 years
- **(E)** 8 years
- **(F)** 12 years
- **(G)** 5 years

79. Citric acid is 6% by mass of a fruit drink. A glass holds 200 g of the drink. What mass of citric acid is in it? [Acids, Bases & Salts]

- **(A)** 6 g
- **(B)** 1.2 g
- **(C)** 60 g
- **(D)** 12 g
- **(E)** 120 g
- **(F)** 18 g
- **(G)** 24 g

80. Write 3,40,00,000 (three crore forty lakh) in standard form. [Exponents & Powers]

- **(A)** 34 x 10^6
- **(B)** 3.4 x 10^8
- **(C)** 34 x 10^7
- **(D)** 3.4 x 10^7
- **(E)** 3.4 x 10^5
- **(F)** 3.4 x 10^6

81. Water in a xylem vessel rises at 4 mm per second. How far, in metres, does it travel in 1 hour? [Transportation in Animals & Plants]

- **(A)** 1.44 m
- **(B)** 240 m
- **(C)** 14.4 m
- **(D)** 14400 mm
- **(E)** 144 m
- **(F)** 1440 m
- **(G)** 14400 m

82. Twice the sum of a number and 3 gives 22. What is the number? [Simple Equations]

- **(A)** 8
- **(B)** 16
- **(C)** 5
- **(D)** 11
- **(E)** 14
- **(F)** 4
- **(G)** 19

83. Two acidic samples have pH 3 and pH 6; the lower the pH, the stronger the acid. By how many pH units, and which way, is the first stronger? [Acids, Bases & Salts]

- **(A)** 3 units stronger
- **(B)** 9 units stronger
- **(C)** 2 units stronger
- **(D)** 6 units stronger
- **(E)** 1 unit stronger
- **(F)** 18 units stronger

84. Compare 3^3 and 4^2 and state which is larger. [Exponents & Powers]

- **(A)** 4^2 larger by 11
- **(B)** 3^3, smaller by 11
- **(C)** 4^2 larger by 27
- **(D)** 3^3 is larger
- **(E)** 4^2 is larger
- **(F)** both equal 12
- **(G)** they are equal

85. A frog's heart has 3 chambers and a fish's heart has 2. Counting 4 frogs and 5 fish, how many heart chambers are there altogether? [Transportation in Animals & Plants]

- **(A)** 20 chambers
- **(B)** 30 chambers
- **(C)** 9 chambers
- **(D)** 22 chambers
- **(E)** 14 chambers
- **(F)** 18 chambers

86. In a fraction, the denominator is 3 more than the numerator, and the fraction equals 2/3. What is the numerator? [Simple Equations]

- **(A)** 5
- **(B)** 3
- **(C)** 12
- **(D)** 6
- **(E)** 4
- **(F)** 9

87. A cleaner needs pH 10 but the soapy mix reads pH 8. Each cap of soap raises the pH by 0.25. How many caps are needed? [Acids, Bases & Salts]

- **(A)** 4 caps
- **(B)** 16 caps
- **(C)** 8 caps
- **(D)** 10 caps
- **(E)** 5 caps
- **(F)** 0.5 caps
- **(G)** 2 caps

88. Simplify 6^4 / 6^2 to a single number. [Exponents & Powers]

- **(A)** 18
- **(B)** 1296
- **(C)** 216
- **(D)** 1
- **(E)** 12
- **(F)** 6
- **(G)** 36

89. Sweat glands release about 0.8 L of sweat on a hot day. Over a 6-day heat wave, how many litres of water are lost as sweat? [Transportation in Animals & Plants]

- **(A)** 0.48 L
- **(B)** 5.6 L
- **(C)** 4.8 mL
- **(D)** 6.8 L
- **(E)** 4.8 L
- **(F)** 48 L

90. A square has the same perimeter as a rectangle of 10 cm by 6 cm. What is the side of the square? [Simple Equations]

- **(A)** 32 cm
- **(B)** 60 cm
- **(C)** 6 cm
- **(D)** 4 cm
- **(E)** 16 cm
- **(F)** 10 cm
- **(G)** 8 cm

91. Lime water turns milky with carbon dioxide; 1 g of it can detect up to 5 mL of the gas. How much carbon dioxide can 9 g detect? [Acids, Bases & Salts]

- **(A)** 4 mL
- **(B)** 14 mL
- **(C)** 1.8 mL
- **(D)** 45 g
- **(E)** 45 mL
- **(F)** 90 mL
- **(G)** 50 mL

92. Find the value of 2^3 + 3^2 + 1^5. [Exponents & Powers]

- **(A)** 20
- **(B)** 11
- **(C)** 16
- **(D)** 19
- **(E)** 17
- **(F)** 6
- **(G)** 18

93. Eight capillaries side by side equal the width of one human hair, 0.08 mm. What is the width of a single capillary, in millimetres? [Transportation in Animals & Plants]

- **(A)** 0.01 mm
- **(B)** 0.001 mm
- **(C)** 0.16 mm
- **(D)** 0.08 mm
- **(E)** 0.1 mm
- **(F)** 1 mm
- **(G)** 0.64 mm

94. Three times a number exceeds 40 by the same amount that it falls short of 80. What is the number? [Simple Equations]

- **(A)** 20
- **(B)** 10
- **(C)** 120
- **(D)** 30
- **(E)** 40
- **(F)** 60
- **(G)** 12

95. Car-battery acid is 38% sulphuric acid by mass. A battery holds 1500 g of acid solution. What mass is pure sulphuric acid? [Acids, Bases & Salts]

(A) 570 g
(B) 930 g
(C) 38 g
(D) 1140 g
(E) 600 g
(F) 57 g
(G) 380 g

96. If 2 raised to the power x equals 64, what is x? [Exponents & Powers]

(A) 32
(B) 8
(C) 5
(D) 7
(E) 16
(F) 4
(G) 6

97. The lymph system slowly returns about 3 L of fluid to the blood each day. Across the 30 days of June, how many litres is that? [Transportation in Animals & Plants]

(A) 90 mL
(B) 900 L
(C) 60 L
(D) 90 L
(E) 9 L
(F) 10 L
(G) 33 L

98. Half of a number plus one-third of it equals 25. What is the number? [Simple Equations]

(A) 60
(B) 150
(C) 30
(D) 12
(E) 15
(F) 25
(G) 6

99. A neutral salt needs equal-ratio base. A 20 mL acid sample reaches pH 7 with base in a 1 : 1 ratio. How much base does 50 mL of the same acid need for a neutral salt? [Acids, Bases & Salts]

(A) 30 mL
(B) 100 mL
(C) 45 mL
(D) 70 mL
(E) 25 mL
(F) 20 mL
(G) 50 mL

100. Simplify (10^3) raised to the power 2. [Exponents & Powers]

(A) 10^5

(B) 10^8

(C) 10^6

(D) 10^9

(E) 100^3

(F) 10^3